3.5 Exercises

3.5.1. (a)
$$\begin{aligned} 3x &\equiv 2 && \pmod 7 \\ x &\equiv 2\cdot(3^{-1}) && \pmod 7 \\ &\equiv 2\cdot 5 && \pmod 7 \\ &\equiv 10 && \pmod 7 \\ &\equiv 3 && \pmod 7 \end{aligned}$$

Note that $3\cdot 5 = 15 \equiv 1 \pmod 7$

Therefore, $x \in \{3 + 7\mathbb{Z}\}$

(b)
$$\begin{aligned} 5x+1 &\equiv 13 && \pmod{13} \\ 5x &\equiv 12 && \pmod{13} \\ x &\equiv 12\cdot(5^{-1}) && \pmod{13} \\ x &\equiv 12\cdot 8 && \pmod{13} \\ x &\equiv 96 && \pmod{13} \\ x &\equiv 5 && \pmod{13} \end{aligned}$$

Note that $5\cdot 8 = 40 \equiv 1 \pmod{13}$

Therefore, $x \in \{5 + 13\mathbb{Z}\}$

(c)
(d)
(e) TO DO
(f)

3.5.2. a) NOT A GROUP:
- 1st row: $aa = a$, $ad = a$; cancellation implies $a = d$.
- Since $a \neq d$ in G, we reach a contradiction.

d) NOT A GROUP:
- 4th row: $da = d$, $db = d$; cancellation implies $a = b$.
- Since $a \neq b$ in G, we reach a contradiction.

(b) GROUP:
- Table is Latin Square
- Row/Column 1 $\Rightarrow$ identity element is: a
- $bb = a$, $cc = a$, $dd = a$; each element is own inverse (order 2)
- Shares multiplication table of V_4.

c) GROUP:
- Table is Latin Square
- Row/Column 1 $\Rightarrow$ identity element is: a
- $cc = a$, $bb = c$, and $dd = c$ $\Rightarrow$ orders of b, c, d are $4, 2, 4$ respectively.
- Shares multiplication table of C_4.

3.5.3. CAYLEY TABLES

- Symmetries of Rectangle

$\circ$	id	r	v	h
id	id	r	v	h
r	r	id	h	v
v	v	h	id	r
h	h	v	r	id

- $(\mathbb{Z}_4, +)$

$\circ$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

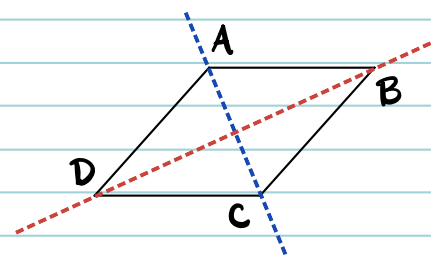
Let r := 180° rotation,
v := vertical axis reflection
h := horizontal axis reflection

There are 4 elements in each group, but the groups are not the same.

In symmetries of rectangles, all non-id elements have order 2.
In $(\mathbb{Z}_4, +)$, two elements have order 4 (1 and 3), one element order 2 (2).

3.5.4.

Symmetries of rhombus are
- identity, id
- 180° rotation, r
- diagonal reflection #1 (top left --- bottom right), d_1
- diagonal reflection #2 (bottom left --- top right), d_2



- Symmetries of Rectangle

$\circ$	id	r	v	h
id	id	r	v	h
r	r	id	h	v
v	v	h	id	r
h	h	v	r	id

- Symmetries of Rhombus

$\circ$	id	r	d_1	d_2
id	id	r	d_1	d_2
r	r	id	d_2	d_1
d_1	d_1	d_2	id	r
d_2	d_2	d_1	r	id

STRUCTURALLY THE SAME...

3.5.5. TODO